A real-machine reference

Bent box boom. Compact pivots. Inward bucket. Four resized V6 actuators.

Cat 320 manufacturer reference



1:12 V6 model — arm and fixed mount



Reference image © Caterpillar. Boom / stick / bucket width / tip radius follow the published 1:12 dimensions.

Published Cat 320 dimension	Reference	Model at 1:12
Reach boom pin centres	5700	475.000
R2.9 stick pin centres	2900	241.667
1.19 m3 HD bucket width	1200	100.000
Bucket tip radius	1570	130.833

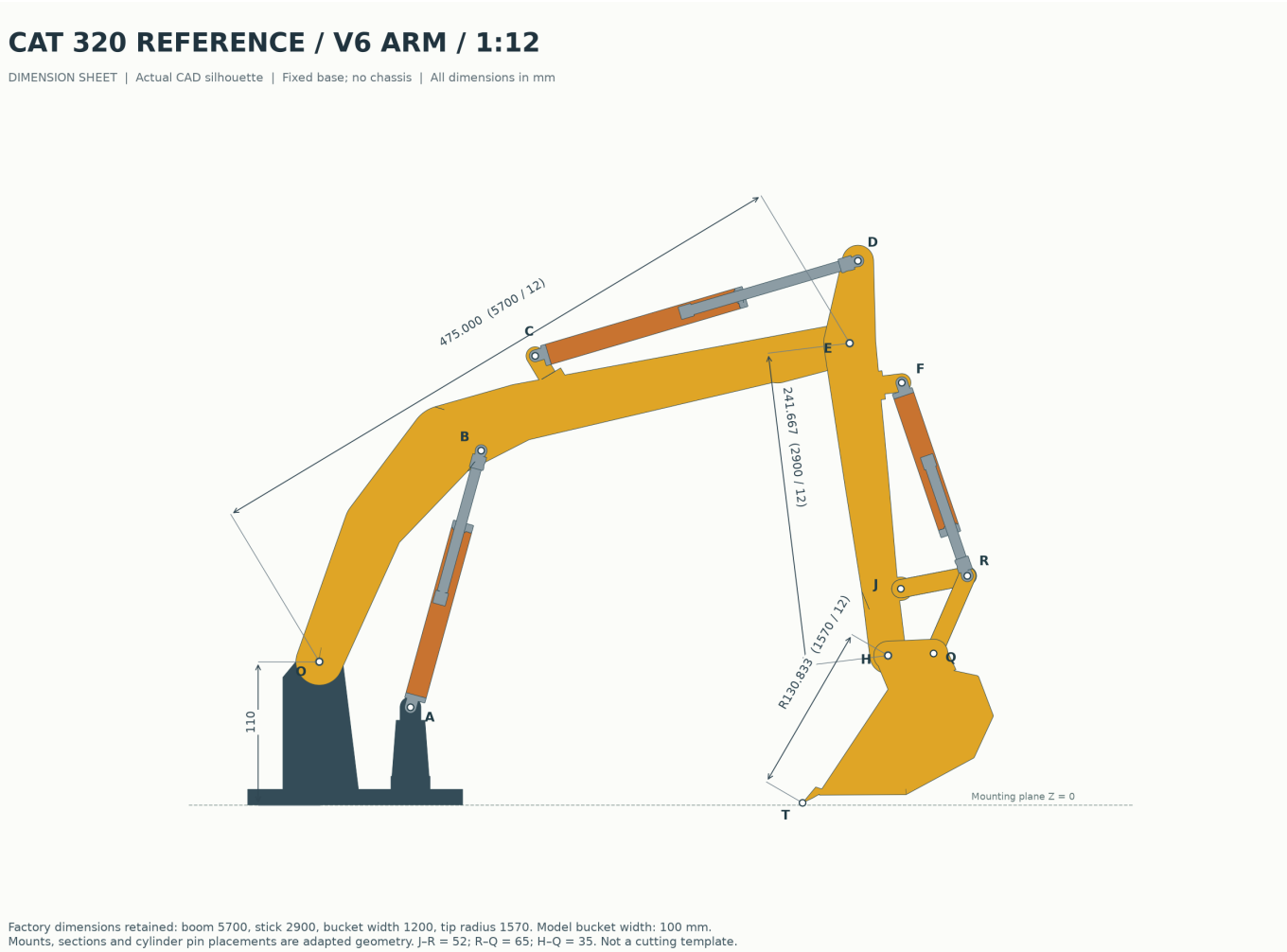
Confirmed dimensions and adapted geometry

The four dimensions above come from Caterpillar technical specifications, pages 4, 5 and 11. Side photographs establish the shape and mechanism. Section outlines, detailed pin locations and the ground support are adapted to the V6 actuators; this is not factory CAD or a scaled factory load rating.

References: Finning Cat 320 product page; Caterpillar specifications and 360-degree side image. Full URLs and downloaded references are included under references/. Manufacturer images remain copyright Caterpillar.

Geometry & correct bucket action

Side silhouette is generated from the exported CAD meshes.



Factory dimensions retained: boom 5700, stick 2900, bucket width 1200, tip radius 1570. Model bucket width: 100 mm.
Mounts, sections and cylinder pin placements are adapted geometry. J–R = 52; R–Q = 65; H–Q = 35. Not a cutting template.

Key interface	Model dimensions
Main pivots O, E, H	diameter 6 shaft / diameter 6.2 hole
Actuator and linkage pivots	diameter 3 shaft / diameter 3.2 hole
Four-bar J-R / R-Q / H-Q	52 / 65 / 35
Boom / stick box width	48 / 32
Base plate / root height	165 x 164 x 12 / 110

Extension must produce these actions

Boom rises. Stick crowds inward. Bucket cutting lip curls up toward the base.
Both boom cylinders share one angle and must remain equal in length.
The bucket mouth faces the machine in the reference pose. Its tip lies
approximately 1.6 mm above the mounting plane in that pose.

Actuators sized to the arm

The V6 cross-section and screw drive are retained; the axial lengths change.

Function	Qty	Screw	Stroke	Closed-open
Boom, V6-B	2	100	85	165-250
Stick, V6-T	1	120	105	185-290
Bucket, V6-K	1	75	60	140-200

Screw length is measured from gearbox front face to tip.
V6 closed pin spacing = screw length + 65; stroke = screw length - 15.
Each end retains 3 mm of mechanical reserve beyond nominal travel.
The original 55 mm V6-S screw and its PCB do not fit these longer variants.

Matching regenerated components

Each variant has matching barrel, front cap, rear cap, rod and rod eye files.
New end-stop boards, Gerbers, KiCad, BOM and placement files are included.
All three actuator CAD/STL checks pass; all three boards have zero design-rule errors and zero independent Gerber readback mismatch.

Rotation and electrical direction

M3 x 0.5 right-hand screw: 2 revolutions per mm.
Clockwise extends, viewed from the rod end toward the motor.
Nominal full strokes need 170 / 210 / 120 revolutions respectively.
Electrical A+ / B- extends; reverse polarity retracts. Nominal supply is 6 V.
There is no external position feedback from the original V6 circuitry.

Paired boom control

Use three reversible control functions; coordinate both B cylinders together.
They represent one mechanical axis. Do not command unequal lengths.
Match both physical end stops before operation. The viewer sliders specify exact geometry; they are not a supplied hardware position controller.

The original electronics topology and magnetic assumptions are inherited.
The resized mechanical and PCB layouts were checked; powered operation, motor thrust capacity, temperature and real switch positions remain untested.

Assembly & verification

Use the matching sets and check the physical fits before loaded operation.

1. Anchor the fixed base using four M6 bolts through its diameter 6.6 holes.
2. Install boom O, stick E and bucket H with steel shafts and spacers.
3. Fit the paired rockers on J and outer links on R/Q on the illustrated branch.
4. Fit two B cylinders at A-B, T at C-D and K at F-R. Keep B strokes equal.
5. Fit collars with running endplay; add flexible external cable loops.
6. Jog unloaded and verify polarity, physical endpoints and paired reversal.

Measured checks on this revision

134 sampled CAD poses; 997 distinct solid intersections checked.
No external solid interference at the sampled poses, including reserve corners.
461 transmission samples; maximum pin error below 0.000001 mm.
Minimum four-bar transmission angle: 33.16 degrees including reserve.
8 structural STL files: watertight and single-component; STEP imports pass.
Fresh internal checks pass for the three resized actuators and their boards.

Working envelope

The sampled cutting tip spans X 73.3 to 786.8 mm and
Z -246.2 to 715.6 mm relative to the mounting plane.
Below-plane poses represent excavation. A solid-floor rig must restrict these poses or provide a clear recess/bench edge. Sampled checks do not prove collision freedom at every continuous position.

Manufacturing and load limits

The boom STL is 509 mm long; it needs a large printer or an engineered split.
Closed hollow boxes require process-specific support and cavity planning.
A watertight model is not a validated load-bearing printed assembly.
Actual collar set screws, external wires and floor material are outside the collision model. Printing, alignment, rod deflection, PETG creep, gearbox thrust capacity and real end-stop accuracy need physical prototype tests.
No payload rating is supplied. See README.md for the shaft/spacer BOM.